MA 324
STATISTICAL INFERENCE AND MULTIVARIATE ANALYSIS
IIT GUWAHATI

END-SEMESTER EXAMINATION
14:00–17:00 IST
MAY 8, 2026

PART-B

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Signature of the Invigilator:

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	3	3	0	1	07

QUESTIONS

1. (6 points) Let

$$\boldsymbol{S} = \begin{pmatrix} S_{yy} & S_{y\boldsymbol{X}}^{T} \\ S_{y\boldsymbol{X}} & S_{\boldsymbol{XX}} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1\times 1}$ and $\boldsymbol{X}_{q\times 1}$ based on a centered (with respect to sample mean) sample of size n. Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector $\boldsymbol{X}$.

r_1^2 is the ~~eigenvector~~ eigenvalue of $\cancel{S_{yy}^{-1/2} S_{12} S_{22}^{-1}}$ $S_{yy}^{-1/2} S_{yx} S_{xx}^{-1} S_{xy} S_{yy}^{-1/2}$

Now, since Y is 1×1, this is equivalent to:

$$\frac{1}{SS_T}\left(S_{yx} S_{xx}^{-1} S_{xy}\right)$$

Here $S_{yx} = [Cov(Y, X_1) \quad Cov(Y, X_2) \quad \dots \quad Cov(Y, X_q)]$

and $S_{xy} = S_{yx}^{T}$

then

$$S_{yx} S_{xx}^{-1} S_{yx}^{T} = [Cov(Y,X_1) \dots Cov(Y,X_n)]\, S_{xx}^{-1} \begin{bmatrix} Cov(Y,X_1) \\ \vdots \\ Cov(Y,X_n) \end{bmatrix}$$

$$= \left[\sum_{i=1}^{q} (Cov(Y, X_i))^2 \times \left(\begin{matrix}\text{Sum of } i\text{th} \\ \text{column of } S_{xx}^{-1}\end{matrix}\right)\right]_{1\times 1}$$

Now, the i^{th} column of S_{xx}^{-1} adds up to $\frac{S_{xx}}{(x_i - \bar{x})^2}$

then the matrix is:

$$\left[\sum_{i=1}^{q} \frac{(y_i - \bar{y})^2 (x_i - \bar{x})^2 (x_i - \bar{x})^2}{S_{xx}}\right]$$

Now importantly for a regression through origin, i.e having transferred $y_i \to y_i - \bar{y} = y_i^*$ and $x_i \to x_i - \bar{x} = x_i^*$

we get $\hat{\beta}_1 = \dfrac{S_{x^* y^*}}{S_{x^* x^*}}$ and $\hat{\beta}_0 = 0$ X.

Then our matrix can be written as

$$S_{yx} S_{xx}^{-1} S_{yx}^T = \left[\sum_{i=1}^{n} (\hat{\beta}_1)^2 (x_i - \bar{x})^2 \right]$$

also $y_i^* = \beta_1 x_i^* \Rightarrow y_i^* = \beta_1 (x_i - \bar{x})$

~~∴ our term is equal to~~ $\cancel{\sum (y^* - \ldots)^2}$ ~~which is equal to~~ $\cancel{\sum (y - \hat{y})^2}$ ~~which is~~ $\cancel{SS_{Reg}}$

∴ Our term is equal to $\sum (\hat{y}^*)^2$

which is equal to $\sum (\hat{y}^* - \bar{y}^*)^2 \because \bar{y}^* = 0$.

which is SS_{Reg} for y^*

~~Converting both to y^*~~ ∵ SS_T would be same for y^* as y

then $r_1^2 = \dfrac{SS_{Reg}}{SS_T}$ for y^* i.e. R^2 for y^*.

∵ Regression ~~Regression~~ on y^* and y will explain the same amount of variance, the R^2 must be equal.

03

$r_1^2 = R^2$ for a multiple linear regression of Y on X.

2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix $\mathbf{S}$. The eigenvalues of $\mathbf{S}$ are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and $\mathbf{e}_4$. You decide to extract $m = 2$ factors using the principal component method.

   (a) (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{\mathbf{L}}$ in terms of the given eigenvalues and eigenvectors.

   (b) (1 point) Let the first row of $\tilde{\mathbf{L}}$ be $\tilde{\mathbf{l}}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.

   (c) (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^{p} \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of $\mathbf{S}$.

(b) $\tilde{h}_1^2 = (0.8)^2 + (-0.4)^2 = 0.64 + 0.16 = 0.80$

01

(a) The Loading matrix $\tilde{L}$ will be

$$\cancel{\tilde{L} = \begin{bmatrix} e_{11}\sqrt{\lambda_1} & e_{21}\sqrt{\lambda_2} \\ e_{12}\sqrt{\lambda_1} & e_{22}\sqrt{\lambda_2} \\ e_{13}\sqrt{\lambda_1} & e_{23}\sqrt{\lambda_2} \\ e_{14}\sqrt{\lambda_1} & \end{bmatrix}}$$

0

$$\tilde{L} = \begin{bmatrix} e_{11}\sqrt{\lambda_1} & e_{21}\sqrt{\lambda_2} \\ e_{12}\sqrt{\lambda_1} & e_{22}\sqrt{\lambda_2} \\ e_{13}\sqrt{\lambda_1} & e_{23}\sqrt{\lambda_2} \\ e_{14}\sqrt{\lambda_1} & e_{24}\sqrt{\lambda_2} \end{bmatrix} T_{2\times 2}$$

where T could be any 2×2 orthogonal matrix (i.e. $TT^T = I$)

(c) $\sum_{i=1}^{p} h_i^2$ = sum of squares of all elements of L.

To get this, we calculate $\tilde{L}L^T$ and take its trace

This is
$$\begin{bmatrix} e_{11}\sqrt{\lambda_1} & e_{21}\sqrt{\lambda_2} \\ e_{12}\sqrt{\lambda_1} & e_{22}\sqrt{\lambda_2} \\ e_{13}\sqrt{\lambda_1} & e_{23}\sqrt{\lambda_2} \\ e_{14}\sqrt{\lambda_1} & e_{24}\sqrt{\lambda_2} \end{bmatrix} T^T \; T \begin{bmatrix} e_{11}\sqrt{\lambda_1} & e_{12}\sqrt{\lambda_1} & e_{13}\sqrt{\lambda_1} & e_{14}\sqrt{\lambda_1} \\ e_{21}\sqrt{\lambda_2} & e_{22}\sqrt{\lambda_2} & e_{23}\sqrt{\lambda_2} & e_{24}\sqrt{\lambda_2} \end{bmatrix}$$

$\because T^T T = I$

$\therefore$ we get the trace as :

$$e_{11}^2 \lambda_1 + e_{12}^2 \lambda_1 + e_{13}^2 \lambda_1 + e_{14}^2 \lambda_1 + e_{21}^2 \lambda_2 + e_{22}^2 \lambda_2$$
$$+ e_{23}^2 \lambda_2 + e_{24}^2 \lambda_2$$
$$= \lambda_1 + \lambda_2$$

$\therefore$ Sum of all estimated communalities is exactly equal to sum of the largest 2 eigenvalues of S.

02

3. (5 points) Let $X_1, X_2, \ldots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2. Let $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Derive a variance-stabilizing transformation $g(\overline{X}_n)$ such that the asymptotic variance of $g(\overline{X}_n)$ does not depend on μ.

We can take

$g(\bar{X}_n) = \frac{\bar{X}_n - \mu}{\mu/\sqrt{n}}$ which is asymptotically distributed as $N(0,1)$ & and thus its asymptotic variance will not depend on μ.

(by Central Limit Theorem)

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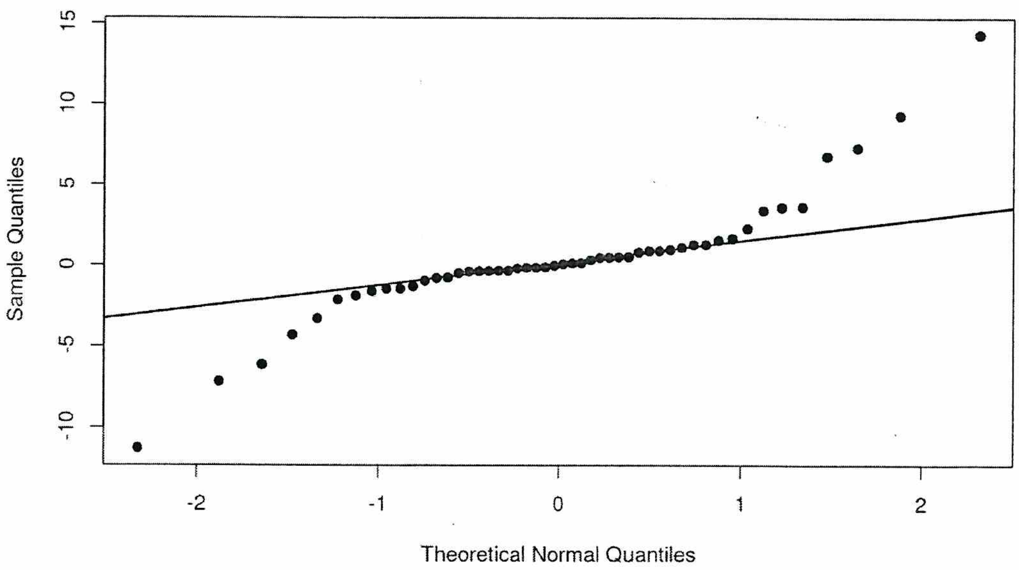
4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



As per the plot, the data does NOT follow a normal distribution. The distribution is skewed. The data can be characterized by lognormal distribution via necessary transformations on the original data.

0 1

Additional Pages for Answers